DATA TYPES IN C

Each variable in C has an associated data type. It specifies the type of data that the variable can store like integer, character, floating, double, etc. Each data type requires different amounts of memory .

```
#include<stdio.h>

int main()
{

	printf("hello\n");
	puts("hello");

}
```

O/P=hello
hello

//

```
#include<stdio.h>

int main()
{

	printf("hello","hai");


}
```

//////////output:hello

//

```
#include<stdio.h>

int main()
{

	puts("hello","hai");//wrong(x)

}
```

//

Print HELLO HAI using printf in one stattement

```
printf(“hello %s”,”hai”);
```

//

```
#include<stdio.h>
int main()
{
printf("%s hello %s ","abc","xyz");////abc hello xyz
}
```

//

Data
Example resume/biodat
name:
age:
dob:
place:
id

//

->CONSTANT
->VARIABLE

character constant->’A’, ‘*’,’#’,’5’

Integer constant->56,-5,0,034,0X12
Real constant->5.6,1234.5,3.4E+2
String constant->”a”, “vector”,”5.67”

//

Variable declaration

datatype var_name;//variable declaration
datatype var_name=data;///variable declaration and initialization

//

DATA TYPES ARE CLASSIFIED AS:

Premitive/basic-> char ,int ,float ...
userdefined:array,strings,struct,union,enum

Baisc Data:
------>Integral:char , short int, int ,long int ,long long int

char=1byte
short int=2byte
int =2/4 bytes
long int=4 or 8 bytes
long long int=8bytes

---------->real:float, double, long double

float=4bytes
double=8 bytes
long double=10/12/16

Variables are identifiers and need to be declared.

//

1:#include<stdio.h>

```
int main()
{
	printf(v);///varaibles are identifiers and need to be declared
	putchar('v');////o/p=v
}
```

character

The **char keyword** is used to declare the variable of character type:

```
char var_name;
```

Format specifier for character

format specifier for char=%c
Format specifier for string=%s

Format specifier for Integer

%d %u
%o=octal
%x=Hex

format specifier for Float
%f %e =float
%lf=double
%LF=long double

//

Type-modifier variable declaration

Type Modifiers are **special keywords defined in the programming language which are used to modify the default properties of the Built-in Data types**.

short
long
signed
unsigned

Character Data Type

Character data type allows its variable to store only a single character. The size of the character is 1 byte. It is the most basic data type in C. It stores a single character and requires a single byte of memory in almost all compilers.

signed char -->sign bit is present at MSB

unsigned char->sign bit is absent

default is signed

char x;////signed char

//
///

char=>1 byte=8 bits
MAXIMUM range is 2^8=256(0–255)

unsigned char-(0 -255)
signed char— (-128 ....0.....+127)

//

//
How negative values are stored?
-1? negative values are stored in 2's complement form

1=0 0 0 0 0 0 0 1
1'S comp=1 1 1 1 1 1 1 0
2's comp=1's comp+1

1 1 1 1 1 1 1 0 (1's comp)
1+
---
1 1 1 1 1 1 1 1

//

unsigned char v=255
1 1 1 1 1 1 1 1
//
Unsigned char var=128
1 0 0 0 0 0 0 0

Signed char var=-128

stored in 2's comp
128=1 0 0 0 0 0 0 0
1's=0 1 1 1 1 1 1 1
2's= 0 1 1 1 1 1 1 1
+ 1
---
1 0 0 0 0 0 0 0

//
```
#include <stdio.h>
int main()
{
  char x=65,y=0;
   y=x+1;
   printf("%d\n",x);
   printf("%c\n",x);
   printf("%d",y);
}
```

2.

```
 char x=127,y=0;
 y=x+1;
printf(“%d”,y);
```

//